Zikun Ouyang

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Research interests

Probablity theory and statistical mechanics.
Keywords: Integrable probability, KPZ universality, directed polymers, random matrix theory, special functions.

Publications

[1] Law of large numbers for directed polymers with matrix disorder, with Guillaume Barraquand and Bartosz Kołodziejek. *In preparation*.

[2] Exponential functionals of two random walks and stationary measures of discrete growth processes. *Preprint*, ArXiv:2610.19673.

[3] Stationary inverse-Wishart polymers, with Guillaume Barraquand. *Preprint*, ArXiv:2511.14375.

[4] Generalized unistochastic matrices, with Ion Nechita and Anna Szczepanek. *Linear Algebra and its Applications*. ArXiv:2310.03436.

Education

- **Université Paris Dauphine-PSL** — *10/2024 - 06/2027 (expected)*
  PhD in Mathematics, advised by Guillaume Barraquand and Cristina Toninelli — Paris, France
- **Université Paris Dauphine-PSL** — *09/2023 - 06/2024*
  Master 2 in Mathematics, with a minor in physics and highest honors — Paris, France
- **Université Paul Sabatier** — *09/2022 - 06/2023*
  Master 1 in Mathematics, with highest honors — Toulouse, France
- **University of Science and Technology of China** — *09/2019 - 06/2023*
  Bachelor of Science in Mathematics, China-France Mathematics Talent Class (CTMCF) program — Hefei, China

Invited talks

- **Integrable Probability, Classical and Quantum Integrability** — *26/05/2026*
  Stationary inverse-Wishart polymers, seminar — Online
- **Young Researcher Seminar** — *06/11/2025*
  Integrable directed polymers and their stationary measures, seminar — Université Paris Dauphine

Contributed talks and posters

- **4th Simons Math Summer Workshop: Algebraic Methods in Probability** — *13/07/2026*
  Stationary inverse-Wishart polymers, lightning talk — Stony Brook University
- **Integrable combinatorics – IMJ-PRG summer school 2026** — *26/06/2026*
  Stationary inverse-Wishart polymers, contributed talk — Sorbonne Université
- **Universality in the Kardar–Parisi–Zhang class and random matrix theory** — *02/06/2026*
  Stationary inverse-Wishart polymers, lightning talk — ICMS, Edinburgh
- **Troisième rencontre de probabilités intégrables** — *14/11/2025*
  Stationary inverse-Wishart polymers, short presentation — Institut Henri Poincaré
- **Random Matrix Theory Summer School in Japan 2025** — *11/09/2025*
  Stationary measures of inverse-Wishart polymer, poster session — Kyoto University

Teaching

- **Tutorial: Discrete processes** — *2026-2027*
  Master 1, Université Paris Dauphine-PSL — in French
- **Tutorial: Statistical Modelling** — *2025-2026*
  Licence 3, Université Paris Dauphine-PSL — in English
- **Tutorial: Introduction aux probabilités** — *2025-2026*
  Licence 2, Université Paris Dauphine-PSL — in French
- **Tutorial: Introduction aux probabilités** — *2024-2025*
  Licence 2, Université Paris Dauphine-PSL — in French

Service

- **École d'automne 2025 GDR MEGA** — *13-17 oct. 2025*
  Co-organized one-week autumn school — Les Plantiers, France

Languages

Chinese (native), English (C1), French (B2).